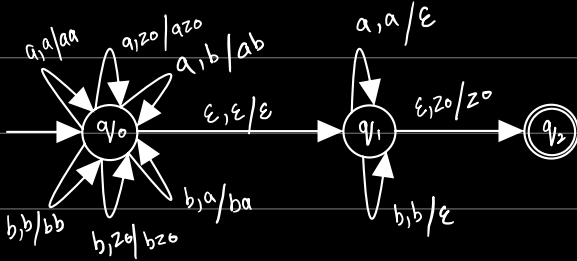


i) $S \rightarrow (s) | ss | \epsilon$

ii) a) $S \rightarrow aSa$

$S \rightarrow bSb$

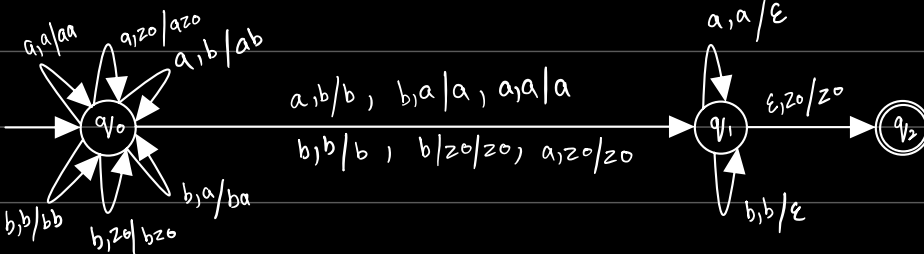
$S \rightarrow \epsilon$



b) $S \rightarrow aSa$

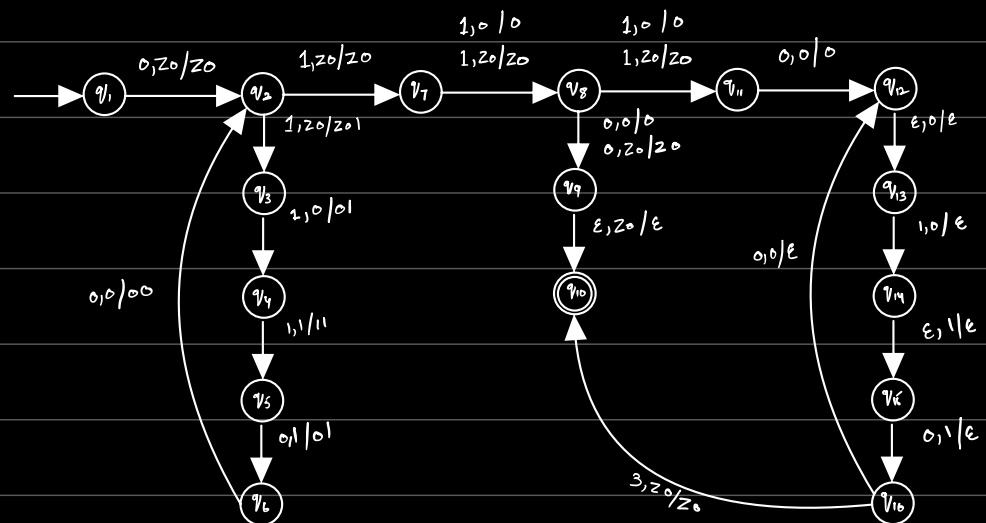
$S \rightarrow bSb$

$S \rightarrow a/b$



iii) $S \rightarrow 01A$

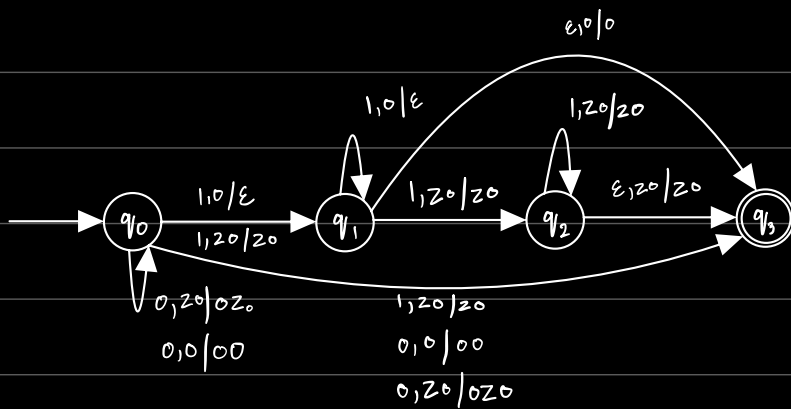
$A \rightarrow 1100A10 | 110$



iv) $S \rightarrow A \mid B$

$A \rightarrow 0A \mid 0A1 \mid 0$

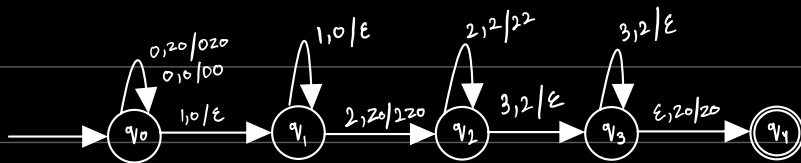
$B \rightarrow 0B1 \mid B1 \mid 1$



v) $S \rightarrow AB$

$A \rightarrow 0A1 \mid 01$

$B \rightarrow 2B3 \mid 23$



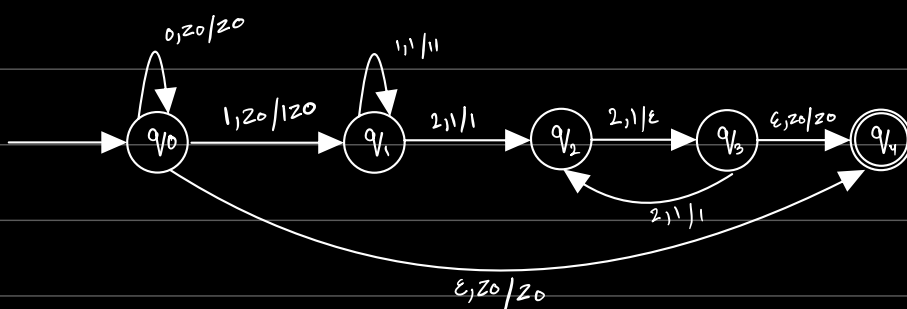
vi) Not Possible bcz the language is not context free.

vii) Not Possible bcz the language is not context free.

viii) $S \rightarrow AB$

$A \rightarrow 0A \mid \epsilon$

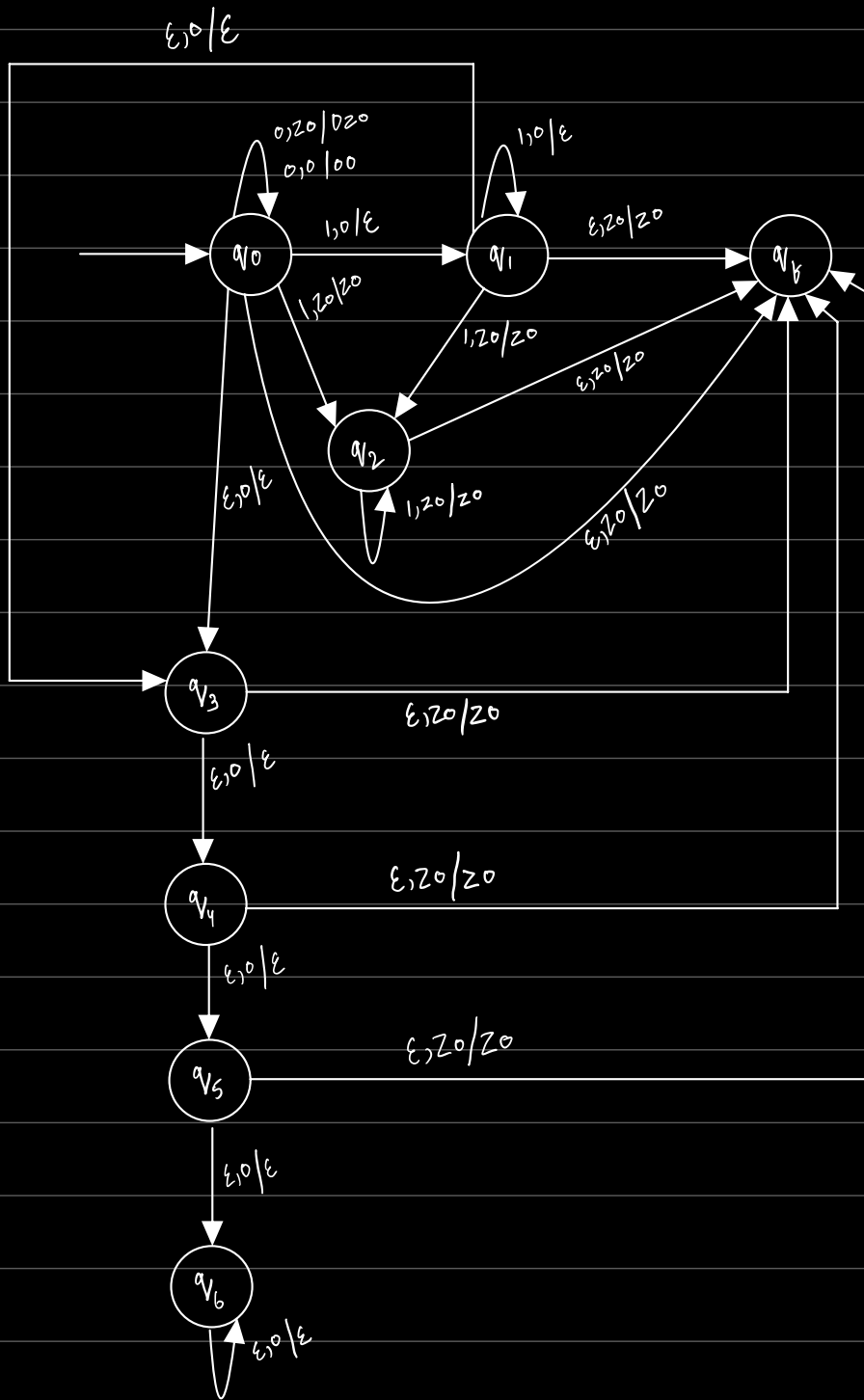
$B \rightarrow 1B22 \mid \epsilon$



IX) $S \rightarrow 0S1 \mid A$

$A \rightarrow B \mid \epsilon \mid 0 \mid 00 \mid 000$

$B \rightarrow 1B \mid \epsilon$



X) Not Possible bcz the language is not context free.

Q2)

$$a) M = \{Q, \Sigma, \Gamma, \delta, q_0, Z_0, F\}$$

$$1) Q = \{q_0, q_1, q_2\}$$

$$2) \Sigma = \{ (,), [,], \{, \} \}$$

$$3) \Gamma = \{ _, \{, \}, Z_0 \}$$

$$4) q_0 = q_0$$

$$5) Z_0 = Z_0$$

$$6) F = \{q_2\}$$

$$7) \text{ let } x \in \Gamma$$

$$\delta(q_0, (, x) = \{q_0, (x)\}$$

$$\delta(q_0, [, x) = \{q_0, [x]\}$$

$$\delta(q_0, \{, x) = \{q_0, \{x\}\}$$

$$\delta(q_1,) = \{q_1, \epsilon\}$$

$$\delta(q_1,] = \{q_1, \epsilon\}$$

$$\delta(q_1, \} = \{q_1, \epsilon\}$$

$$\delta(q_0, \epsilon, Z_0) = \{q_2, Z_0\}$$

$$\delta(q_1, \epsilon, Z_0) = \{q_0, Z_0\}$$

$$\delta(q_0,), \epsilon) = \{q_1, \epsilon\}$$

$$\delta(q_0,], \epsilon) = \{q_1, \epsilon\}$$

$$\delta(q_0, \}, \epsilon) = \{q_1, \epsilon\}$$

$$\delta(q_1, (, x) = \{q_0, (x)\}$$

$$\delta(q_1, [, x) = \{q_0, [x]\}$$

$$\delta(q_1, \{, x) = \{q_0, \{x\}\}$$

b) - When the PDA reads an opening bracket, it pushes that symbol onto the stack.

- When the PDA reads a closing bracket

- It checks the stack. If the top symbol is the matching opening bracket, it pops it off. this cancels out the pair.

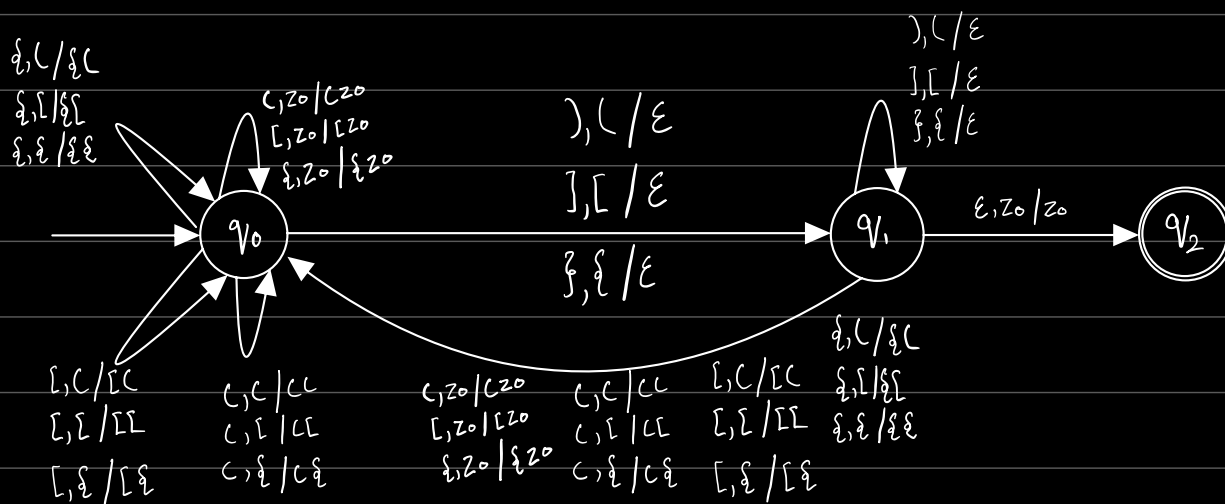
- If the stack top is a mismatch, the PDA rejects the string.

- The PDA accepts string on following conditions

- The entire string has been read.

- The stack is completely empty.

c)



d) Read ' $\{$ ' :

Stack Operation : The PDA is in q_0 , see an opening bracket. It pushes it onto the stack & stays in q_0 .

	$\{$
	Z_0

Read '[' :

Stack Operation : The PDA is in q_0 , see an opening bracket. It pushes it onto the stack & stays in q_0 .

	$[$
	$\{$
	Z_0

Read 'C' :

Stack Operation : The PDA is in q_0 , see an opening bracket. It pushes it onto the stack & stays in q_0 .

	C
	$[$
	$\{$
	Z_0

Read ')':

Stack Operation: The PDA reads ')', sees it matches '(' on the top of stack.

It pops the '(' & moves to state q_1 .

(
{
Z_0

Read 'j':

Stack Operation: The PDA reads 'j', sees it matches 'i' on the top of stack.

It pops the 'i' & stays in state q_1 .

{
Z_0

Read '{':

Stack Operation: The PDA reads '{', sees it matches '}' on the top of stack.

It pops the '}' & stays in state q_1 .

Z_0

— The PDA is in state q_1 , reads no input (ϵ) & sees Z_0 in the stack, moves to q_2 (the final state).

e) $S \rightarrow SS$

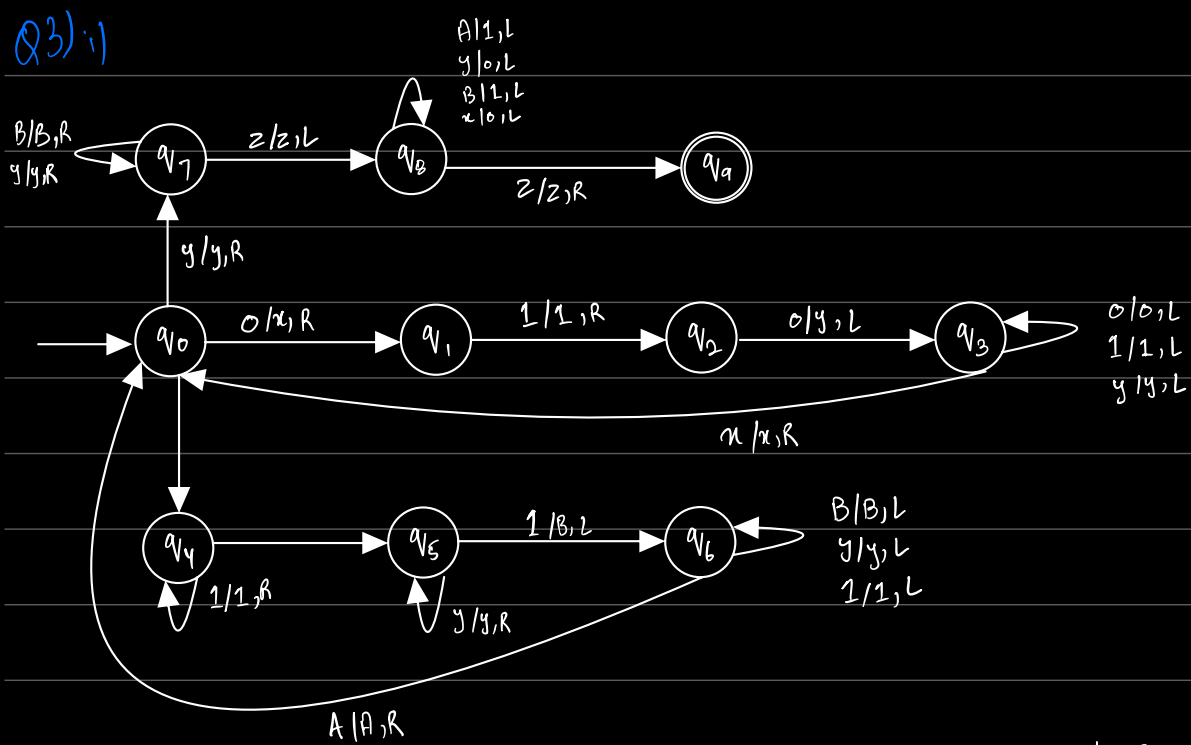
$S \rightarrow (S)$

$S \rightarrow [S]$

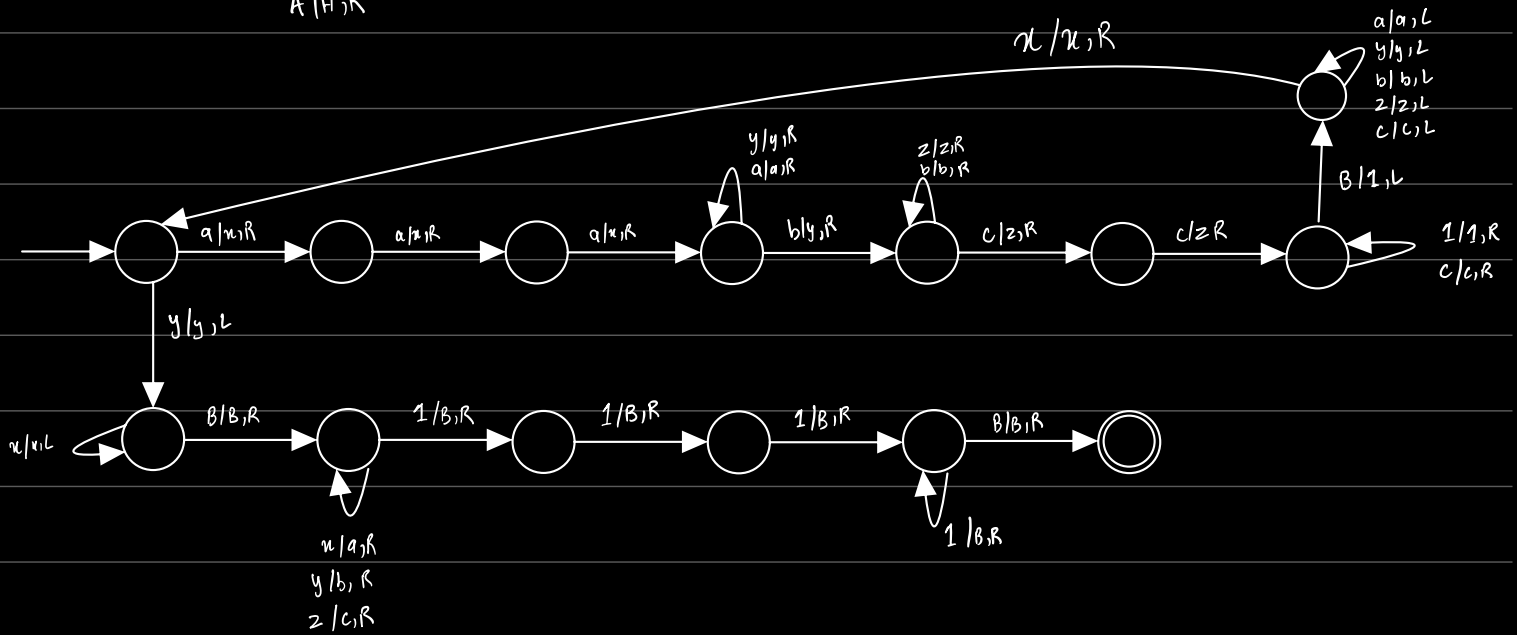
$S \rightarrow \{S\}$

$S \rightarrow \epsilon$

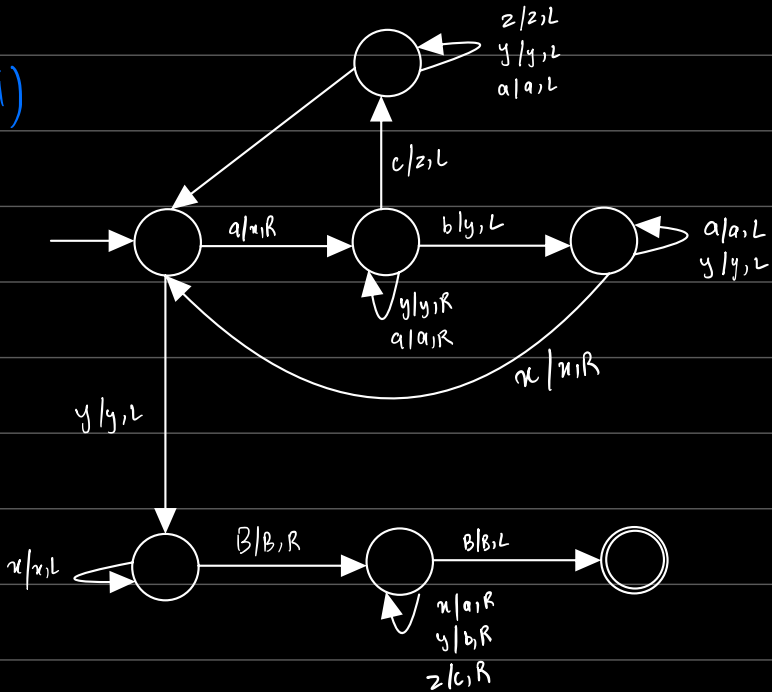
Q3): i)



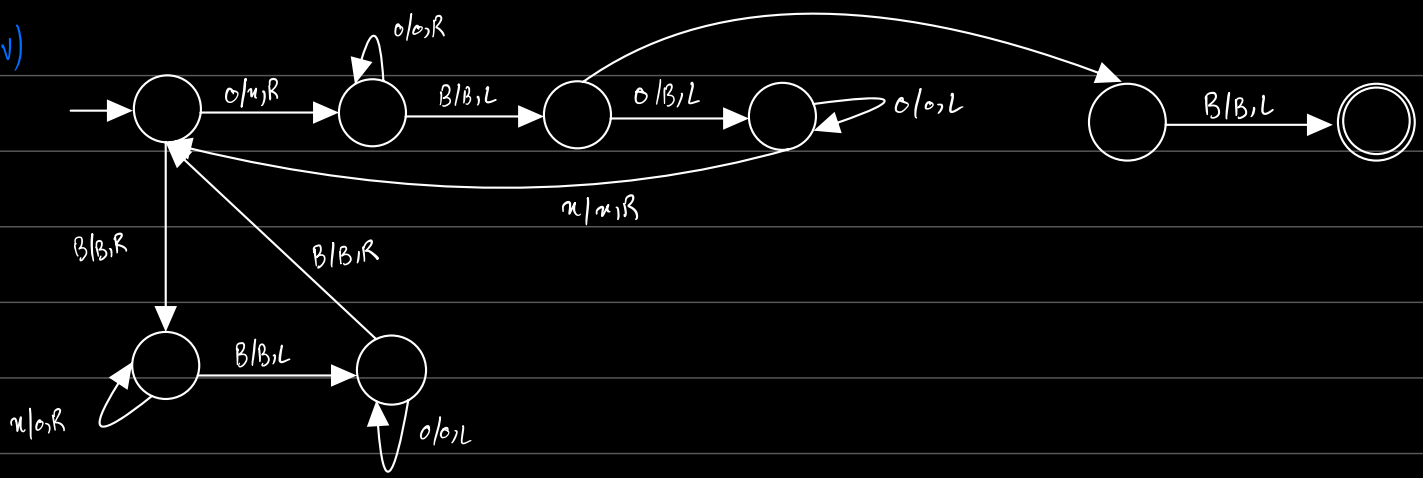
ii)



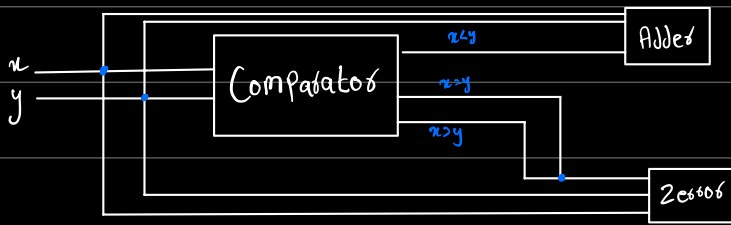
iii)



iv)



v)

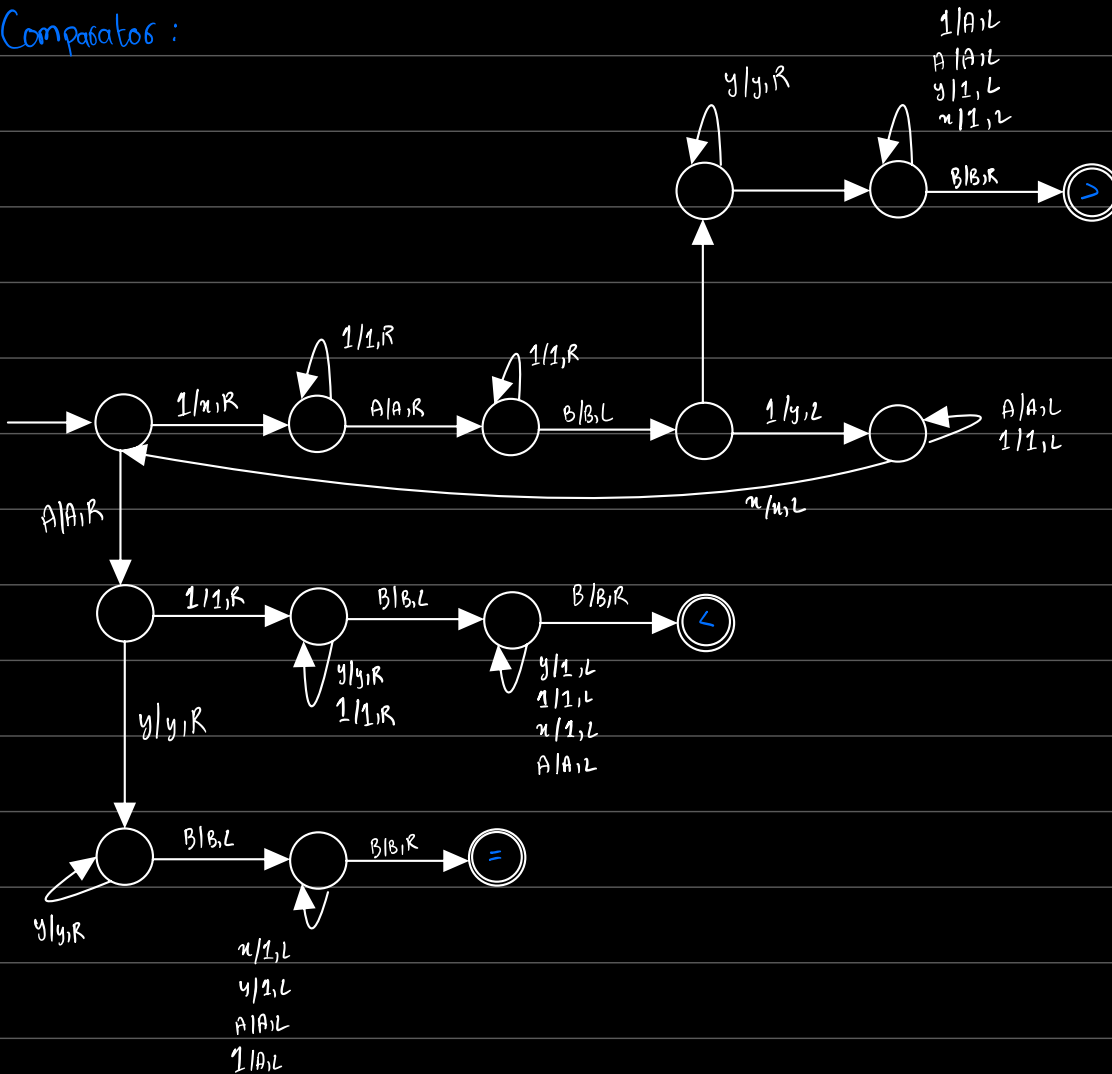


written on tape

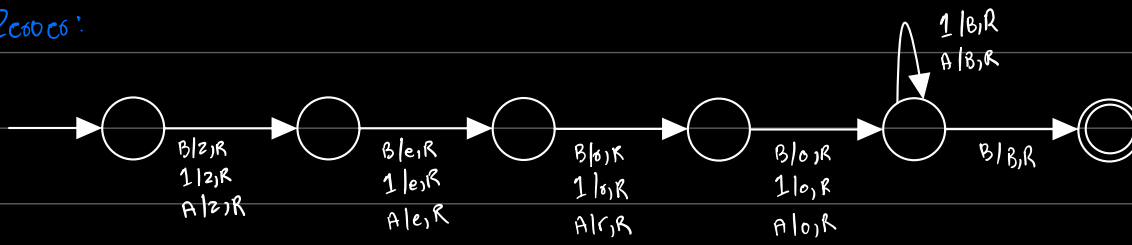
as

x	A	y
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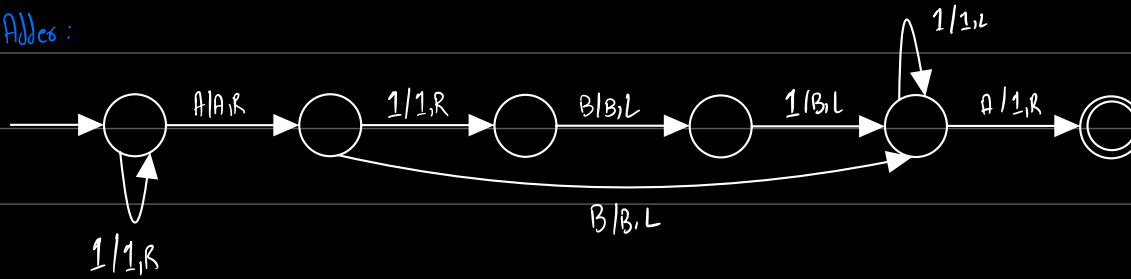
Comparator :



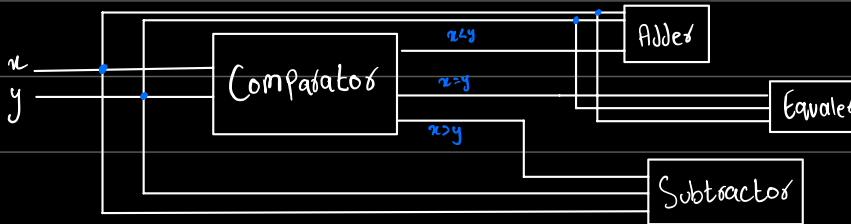
Zeroes:



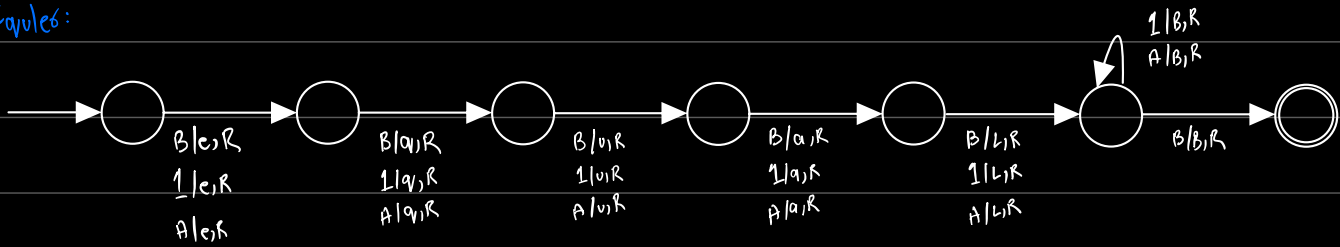
Addes:



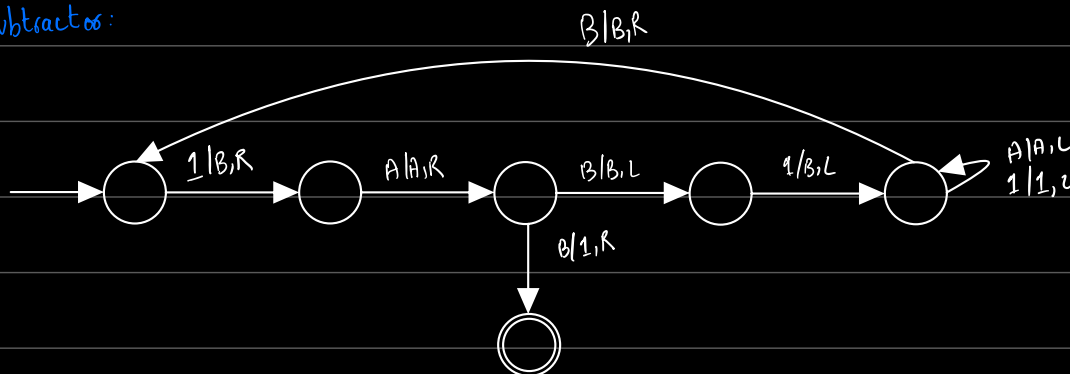
v.)



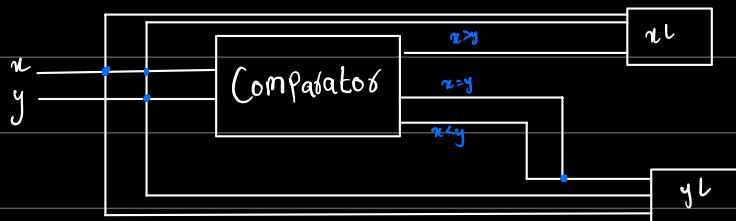
Equaled:



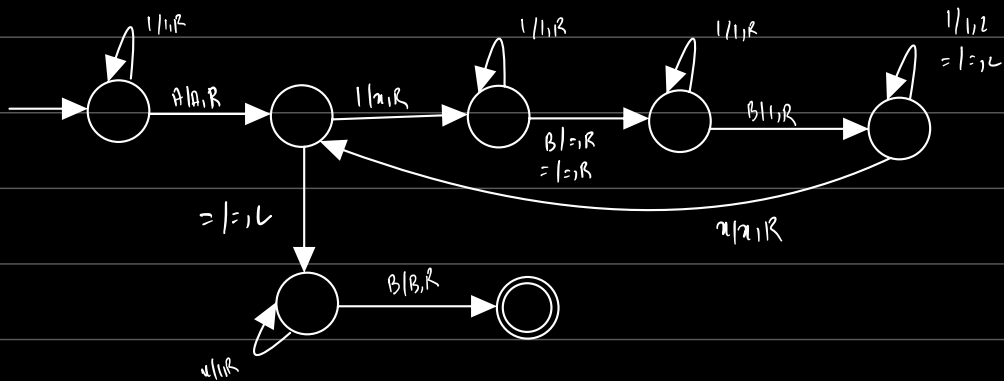
Subtractor:



vii) Just write 64 ones on tape.

$$|x|$$


yz:

 $x)$ 